

# **FAR WESTERN UNIVERSITY**

## **SCHOOL OF ENGINEERING**

**Central Campus**

**Mahendranagar, Kanchanpur**



**A practical Report of :  
GE 245 Engineering Survey II**

**Submitted By :-**

**Name :- Kumar Bhandari**

**Roll No :- Ninrteen '19'**

**Group :- 'D'**

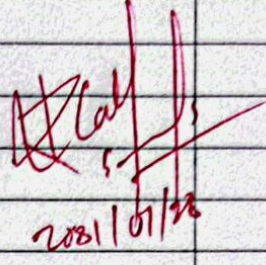
**Semester :- Fourth**

**Symbol No:-**

**Submitted to :-**

**Er. Moti Ram Chaudhary**

# Index

| S.N | Title                                                                                                                 | Date       | Signature                                                                           | Remarks |
|-----|-----------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------|---------|
| 1.  | TRAVERSING                                                                                                            | 2081-01-28 |  |         |
|     | a) To prepare the map of certain area.                                                                                |            |                                                                                     |         |
| 2.  | TACHEOMETRY                                                                                                           | 2081-03-14 |                                                                                     |         |
|     | a) To determine the value of stadia interval factor (k) & stadia constant (C) of the instrument.                      |            |                                                                                     |         |
| 3.  | TRIGONOMETRY LEVELLING                                                                                                | 2081-03-14 |                                                                                     |         |
|     | a) Determination of elevation of point by trigonometric levelling when instrument station are in same vertical plane. |            |                                                                                     |         |
| 4.  | CURVE SETTING                                                                                                         | 2081-03-14 |                                                                                     |         |
|     | a) To set out a simple Circular Curve by Rankine method.                                                              |            |                                                                                     |         |
| 5.  | INDIRECT CONTOURING                                                                                                   | 2081-03-14 |                                                                                     |         |
|     | a) To plot Contour by Square method                                                                                   |            |                                                                                     |         |
|     | b) To prepare a contour map of the given area by radial line method.                                                  |            |                                                                                     |         |

## PRACTICAL NO:1

TITLE :- TRAVERSING

OBJECTIVE :- TO PREPARE THE MAP OF CERTAIN AREA.

INSTRUMENT REQUIRED :-

- i) Pegs
- ii) Ranging rods
- iii) Tape
- iv) Total station
- v) Prism
- vi) Tripod
- vii) Prism stand
- viii) Theodolite

THEORY :-

Traverse means the "literally a line passing through". In Surveying, the linear & angular measurement is done in field with linear & angular measuring instrument. Every activities which are done inside traverse or during traverse is called traversing.

The traversing in which traverse legs are measured by direct chaining on the ground and the traverse angle at every traverse station is measured with theodolite called theodolite traversing. In traversing using total stations, angular as well as linear measurement are measured by total station. The main purpose is to provide control points for different survey and also to provide control points for different and provide alignment of road, canal etc. All the computation of traverse is carried out in tabular form known as Galle's traverse table.

## PROCEDURE:-

### 1. Reconnaissance:-

During reconnaissance, extensive inspection and examination of the entire area to be surveyed is done.

2. Pegs were drawn covering the whole area to be surveyed.

### 3. For measuring angles and distance by total station:-

a) Total station was centered & levelled carefully at station  $D_1$

b) Prism was set out at station  $D_6$  and station  $D_2$

c) Prism of station  $D_2$  was bisected by telescope from face left position and angle was set at zero.

d) Then upper plate was loosen and intersect the centre of prism of station  $D_2$  keeping the face left position.

e) Similarly, both prism of station  $D_2$  and  $D_6$  were bisected from face right position. The average of difference between two reading from face left and two readings from face right position gives the angle at station  $D_1$  from zero set. Same process by setting  $90^\circ$  & average of both gives angle of station  $D_1$ .

f. Similarly, other angles were also measured.

g). To measure distance between two station Prism was bisected at centre and distance (dis) button was pressed. It gives the both horizontal and vertical distance.

Far Western University  
Faculty of Engineering  
University Central Campus

Department of Civil Engineering  
BCE Survey Camp - 2078 : 2078 Batch

Group no: D

Observer: Kumar Bhandari

Recorder: Laxmi Giri

Instrument: Theodolite

5

Bearing: 195°

Date: 2080-12-09

Weather:

Temperature:

Total Station Traverse

Horizontal Angle Observation Sheet

| Inst. and<br>Ht of Inst. | Object<br>Sighted to | Set-I            |     |     |            |      |     | Set-II           |      |     |                  |            |   | Horizontal<br>Distance<br>(m) | Vertical<br>Dist<br>± (m) | Target<br>Ht (m)               | Remarks                        |            |   |   |                  |   |   |
|--------------------------|----------------------|------------------|-----|-----|------------|------|-----|------------------|------|-----|------------------|------------|---|-------------------------------|---------------------------|--------------------------------|--------------------------------|------------|---|---|------------------|---|---|
|                          |                      | HCR observation: |     |     | Face Angle |      |     | Horizontal Angle |      |     | HCR observation: |            |   |                               |                           |                                |                                | Face Angle |   |   | Horizontal Angle |   |   |
|                          |                      | d                | m   | s   | d-m-s      | d    | m   | s                | d    | m   | s                | d-m-s      | d |                               |                           |                                |                                | m          | s | d | m                | s | d |
| D <sub>1</sub>           | D <sub>6</sub>       | 0°               | 0'  | 0"  | 143°33'12" |      |     |                  | 90°  | 0'  | 0"               | 143°32'29" |   |                               | 81.456                    |                                | D <sub>1</sub> →D <sub>2</sub> |            |   |   |                  |   |   |
|                          | D <sub>2</sub>       | 143°             | 33' | 12" |            | 143° | 33' | 4"               | 233° | 32' | 29"              |            |   | 81.452                        |                           | D <sub>2</sub> →D <sub>1</sub> |                                |            |   |   |                  |   |   |
|                          | D <sub>6</sub>       | 180°             | 0'  | 11" | 143°33'56" |      |     |                  | 270° | 0'  | 06"              | 143°32'23" |   |                               |                           |                                |                                |            |   |   |                  |   |   |
|                          | D <sub>2</sub>       | 323°             | 32' | 56" |            |      |     |                  | 53°  | 32' | 29"              |            |   |                               |                           |                                |                                |            |   |   |                  |   |   |
| D <sub>2</sub>           | D <sub>1</sub>       | 0°               | 0'  | 0"  | 98°55'12"  |      |     |                  | 89°  | 59' | 58"              | 98°55'46"  |   |                               | 71.568                    |                                | D <sub>2</sub> →D <sub>3</sub> |            |   |   |                  |   |   |
|                          | D <sub>3</sub>       | 98°              | 55' | 12" |            | 98°  | 55' | 8°5"             | 88°  | 55' | 44"              |            |   | 71.574                        |                           | D <sub>3</sub> →D <sub>2</sub> |                                |            |   |   |                  |   |   |
|                          | D <sub>1</sub>       | 180°             | 0'  | 19" | 98°55'5"   |      |     |                  | 270° | 0'  | 07"              | 98°55'37"  |   |                               |                           |                                |                                |            |   |   |                  |   |   |
|                          | D <sub>3</sub>       | 238°             | 55' | 24" |            |      |     |                  | 8°   | 55' | 24"              |            |   |                               |                           |                                |                                |            |   |   |                  |   |   |
| D <sub>3</sub>           | D <sub>2</sub>       | 0°               | 0'  | 0"  | 128°21'33" |      |     |                  | 90°  | 0'  | 0"               | 128°22'12" |   |                               | 74.525                    |                                | D <sub>3</sub> →D <sub>4</sub> |            |   |   |                  |   |   |
|                          | D <sub>4</sub>       | 128°             | 21' | 33" |            | 128° | 21' | 34"              | 218° | 22' | 12"              |            |   | 74.509                        |                           | D <sub>4</sub> →D <sub>3</sub> |                                |            |   |   |                  |   |   |
|                          | D <sub>2</sub>       | 180°             | 0'  | 0"  | 128°21'35" |      |     |                  | 270° | 0'  | 06"              | 128°21'51" |   |                               |                           |                                |                                |            |   |   |                  |   |   |
|                          | D <sub>4</sub>       | 308°             | 21' | 35" |            |      |     |                  | 38°  | 21' | 57"              |            |   |                               |                           |                                |                                |            |   |   |                  |   |   |
| D <sub>4</sub>           | D <sub>3</sub>       | 0°               | 0'  | 0"  | 126°47'02" |      |     |                  | 90°  | 0'  | 0"               | 126°48'51" |   |                               | 66.252                    |                                | D <sub>4</sub> →D <sub>5</sub> |            |   |   |                  |   |   |
|                          | D <sub>5</sub>       | 126°             | 47' | 02" |            | 126° | 46' | 54°5"            | 216° | 46' | 51"              |            |   | 66.238                        |                           | D <sub>5</sub> →D <sub>4</sub> |                                |            |   |   |                  |   |   |
|                          | D <sub>3</sub>       | 180°             | 0'  | 19" | 126°46'47" |      |     |                  | 270° | 0'  | 18"              | 126°46'29" |   |                               |                           |                                |                                |            |   |   |                  |   |   |
|                          | D <sub>5</sub>       | 306°             | 47' | 06" |            |      |     |                  | 36°  | 46' | 47"              |            |   |                               |                           |                                |                                |            |   |   |                  |   |   |
| D <sub>5</sub>           | D <sub>4</sub>       | 0°               | 0'  | 0"  | 117°57'23" |      |     |                  | 90°  | 0'  | 0"               | 117°57'16" |   |                               | 83.808                    |                                | D <sub>5</sub> →D <sub>6</sub> |            |   |   |                  |   |   |
|                          | D <sub>6</sub>       | 117°             | 57' | 23" |            | 117° | 57' | 22°5"            | 207° | 57' | 17"              |            |   | 83.804                        |                           | D <sub>6</sub> →D <sub>5</sub> |                                |            |   |   |                  |   |   |
|                          | D <sub>4</sub>       | 180°             | 0'  | 57" | 117°57'22" |      |     |                  | 270° | 0'  | 03"              | 117°57'23" |   |                               |                           |                                |                                |            |   |   |                  |   |   |
|                          | D <sub>6</sub>       | 203°             | 57' | 19" |            |      |     |                  | 27°  | 57' | 26"              |            |   |                               |                           |                                |                                |            |   |   |                  |   |   |

Group no: D

Observer: Kumar Bhandari

Recorder: Laxmi Giri

Instrument: Theodolite

Bearing: 195°

Date: 2080-12-09

Weather:

Temperature:

Total Station Traverse  
Horizontal Angle Observation Sheet

| Inst and<br>Ht of Inst | Object<br>Sighted to | Set-I           |     |     |            |     |       | Set-II          |     |     |            |     |     | Horizontal<br>Distance<br>(m) | Vertical<br>Dist<br>±(m) | Target<br>Ht (m) | Remarks |
|------------------------|----------------------|-----------------|-----|-----|------------|-----|-------|-----------------|-----|-----|------------|-----|-----|-------------------------------|--------------------------|------------------|---------|
|                        |                      | HCR observation |     |     | Face Angle |     |       | HCR observation |     |     | Face Angle |     |     |                               |                          |                  |         |
|                        |                      | d'              | m'  | s   | d          | m   | s     | d'              | m   | s   | d          | m   | s   | d'                            | m                        | s                |         |
| D6                     | D5                   | 0°              | 0'  | 0"  | 104°       | 25' | 19.5" | 90°             | 0'  | 0"  | 104°       | 25' | 21" | 68.836                        |                          |                  | D6→D5   |
|                        | D1                   | 104°            | 25' | 21" | 104°       | 25' | 21"   | 194°            | 25' | 19" | 104°       | 25' | 21" | 68.864                        |                          |                  | D1→D6   |
|                        | D5                   | 186°            | 0'  | 01" | 104°       | 25' | 23.5" | 230°            | 0'  | 03" | 104°       | 25' | 28" |                               |                          |                  |         |
|                        | D1                   | 284°            | 25' | 24" | 104°       | 25' | 23.5" | 14°             | 25' | 28" | 104°       | 25' | 28" |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |
|                        |                      |                 |     |     |            |     |       |                 |     |     |            |     |     |                               |                          |                  |         |

### OBSERVATION:-

Observed angle are:

Angles at

$$\text{Station } D_1 = 143^\circ 32' 45''$$

$$\text{Station } D_2 = 98^\circ 55' 25''$$

$$\text{Station } D_3 = 128^\circ 21' 47.75''$$

$$\text{Station } D_4 = 126^\circ 46' 47.25''$$

$$\text{Station } D_5 = 117^\circ 57' 2''$$

$$\text{Station } D_6 = 104^\circ 25' 20.25''$$

### CALCULATION:-

$$(\text{Sum})_{th} = (n-2)180 = (6-2) \times 180 = 720^\circ$$

$$\begin{aligned} (\text{Sum})_{pr} &= 143^\circ 32' 45'' + 98^\circ 55' 25'' + 128^\circ 21' 47.75'' + 126^\circ 46' 47.25'' \\ &\quad + 117^\circ 57' 2'' + 104^\circ 25' 20.25'' \\ &= 719^\circ 59' 26.25'' \end{aligned}$$

$$\begin{aligned} \text{Error} &= (\text{Sum})_{pr} - (\text{Sum})_{th} \\ &= 719^\circ 59' 26.25'' - 720^\circ \\ &= -0^\circ 0' 33.75'' \end{aligned}$$

$$\begin{aligned} \text{Permissible Limit} &= \pm 45'' \sqrt{n} \\ &= \pm 45'' \sqrt{6} \\ &= \pm 1^\circ 50' 13.62'' \end{aligned}$$

So error is within the limit and can be distributed in all stations.

$$\text{So, Correction} = +0^\circ 0' 33.75''$$

$$\begin{aligned} \text{Correction in each station} &= \frac{0^\circ 0' 33.75''}{6} \\ &= 5.63'' \end{aligned}$$

So Corrected angle is

$$\angle D_1 = 143^\circ 32' 45'' + 0^\circ 0' 5.63'' = 143^\circ 32' 50.63''$$

$$\angle D_2 = 98^\circ 55' 25'' + 0^\circ 0' 5.63'' = 98^\circ 55' 30.63''$$

$$\angle D_3 = 128^\circ 21' 47.75'' + 0^\circ 0' 5.63'' = 128^\circ 21' 53.38''$$

$$\angle D_4 = 126^\circ 46' 47.25'' + 0^\circ 0' 5.63'' = 126^\circ 46' 52.88''$$

$$\angle D_5 = 117^\circ 57' 21'' + 0^\circ 0' 5.63'' = 117^\circ 57' 26.63''$$

$$\angle D_6 = 104^\circ 25' 20.25'' + 0^\circ 0' 5.63'' = 104^\circ 25' 25.88''$$

$$\underline{720^\circ 00' 00''}$$

Length measurement:-

length between  $D_1$  &  $D_2$

$$D_1 \rightarrow D_2 = 81.456\text{m}$$

$$D_2 \rightarrow D_1 = 81.452\text{m}$$

$$\text{Average} = \frac{81.456 + 81.452}{2} = 81.454$$

$$\text{Error} = 81.456 - 81.452 = 0.004$$

$$\text{Precision} = \frac{1}{\frac{\text{avg.}}{\text{Error}}} = \frac{1}{\frac{81.454}{0.004}} = \frac{1}{20363}$$

length between  $D_2$  to  $D_3$

$$D_2 \rightarrow D_3 = 71.568$$

$$D_3 \rightarrow D_2 = 71.574$$

$$\text{Average} = \frac{71.568 + 71.574}{2} = 71.571$$

$$\text{Error} = 71.574 - 71.568 = 0.006$$

$$\text{Precision} = \frac{1}{\frac{\text{avg.}}{\text{Error}}} = \frac{1}{\frac{71.571}{0.006}} = \frac{1}{11928.5}$$

Length between  $D_3$  to  $D_4$  :-

$$D_3 \rightarrow D_4 = 74.525$$

$$D_4 \rightarrow D_3 = 74.509$$

$$\text{Avg} = \frac{74.525 + 74.509}{2} = 74.517$$

$$\text{Error} = 74.525 - 74.509 = 0.016$$

$$\text{Precision} = \frac{1}{\frac{\text{Avg.}}{\text{Error}}} = \frac{1}{\frac{74.517}{0.016}} = \frac{1}{4657.312}$$

Length between  $D_4$  to  $D_5$  :-

$$D_4 \rightarrow D_5 = 66.252$$

$$D_5 \rightarrow D_4 = 66.238$$

$$\text{Avg} = \frac{66.252 + 66.238}{2} = 66.3815$$

$$\text{Error} = 66.252 - 66.238 = 0.014$$

$$\text{Precision} = \frac{1}{\frac{\text{Avg}}{\text{Error}}} = \frac{1}{\frac{66.3815}{0.014}} = \frac{1}{4741.53}$$

Length between  $D_5$  to  $D_6$ .

$$D_5 \rightarrow D_6 = 83.808$$

$$D_6 \rightarrow D_5 = 83.804$$

$$\text{Avg} = \frac{83.808 + 83.804}{2} = 83.806$$

$$\text{Error} = 83.808 - 83.804 = 0.004$$

$$\text{Precision} = \frac{1}{\frac{83.806}{0.004}} = \frac{1}{20951.5}$$

length between  $D_6$  to  $D_1$ .

$$D_6 \rightarrow D_1 = 68.876$$

$$D_1 \rightarrow D_6 = 68.864$$

$$\text{Average} = \frac{68.876 + 68.864}{2} = 68.870$$

$$\text{Error} = 68.876 - 68.864 = 0.012$$

$$\text{Precision} = \frac{1}{\frac{\text{Avg}}{\text{Error}}} = \frac{1}{\frac{68.870}{0.012}} = \frac{1}{5739.166}.$$

Here all precision are under permissible limit so,

$$D_1 D_2 = 81.452 \text{ m}$$

$$D_2 D_3 = 71.571 \text{ m}$$

$$D_3 D_4 = 74.517 \text{ m}$$

$$D_4 D_5 = 66.381 \text{ m}$$

$$D_5 D_6 = 83.806 \text{ m}$$

$$D_6 - D_1 = 68.870$$

Now,

$$\text{leg ratio} = \frac{66.381}{83.806}$$

$$= 1:1.26$$

Under permissible limit of 1:3

# GALE'S TRAVERSE TABLE

| Stn            | line                          | length(m)      | Angle           | Bearing         | Consecutive Co-ordinate         |                                  |
|----------------|-------------------------------|----------------|-----------------|-----------------|---------------------------------|----------------------------------|
|                |                               |                |                 |                 | Latitude<br>$L = L \cos \theta$ | Departure<br>$D = L \sin \theta$ |
| D <sub>1</sub> | D <sub>1</sub> D <sub>2</sub> | 81.452         | 143° 32' 50.63" | 195° 0' 0"      | -78.676                         | -21.021                          |
| D <sub>2</sub> | D <sub>2</sub> D <sub>3</sub> | 71.571         | 98° 55' 30.63"  | 113° 55' 30.63" | -29.025                         | 65.421                           |
| D <sub>3</sub> | D <sub>3</sub> D <sub>4</sub> | 74.517         | 128° 21' 53.38" | 62° 17' 24.01"  | 34.65                           | 65.970                           |
| D <sub>4</sub> | D <sub>4</sub> D <sub>5</sub> | 66.381         | 126° 46' 52.88" | 9° 4' 16.89"    | 65.550                          | 10.466                           |
| D <sub>5</sub> | D <sub>5</sub> D <sub>6</sub> | 83.806         | 117° 57' 26.63" | 307° 1' 43.52"  | 50.469                          | -66.905                          |
| D <sub>6</sub> | D <sub>6</sub> D <sub>1</sub> | 68.870         | 104° 25' 25.88" | 231° 27' 9.4"   | -42.917                         | -53.862                          |
|                |                               | $L = 446.597m$ | 720° 0' 0"      |                 | $\Sigma L = 0.051$              | $\Sigma D = 0.009$               |

$$e = \sqrt{(0.051)^2 + (0.009)^2}$$

$$= 0.0517$$

$$\text{Precision Limit} = \frac{1}{P_e}$$

$$= \frac{1}{\frac{446.597}{0.0517}}$$

$$= \frac{1}{8638.24} < \frac{1}{3000}$$

| Correction             |                         | Corrected      |                | Independent Co-ordinate |          |
|------------------------|-------------------------|----------------|----------------|-------------------------|----------|
| Latitude<br>$\Delta L$ | Departure<br>$\Delta D$ | Latitude       | Departure      | L                       | D        |
| -0.0133                | -0.000668               | -78.6893       | -21.0816       | 140                     | 80       |
| -0.00491               | -0.00207                | -29.0299       | 65.4189        | 61.3107                 | 58.9184  |
| -0.00586               | -0.00209                | 34.644         | 65.9679        | 32.2808                 | 124.3373 |
| -0.01109               | -0.000332               | 65.5389        | 10.465         | 66.9248                 | 190.3052 |
| -0.00854               | -0.00212                | 50.46          | -66.907        | 132.4637                | 200.7702 |
| -0.007264              | -0.001708               | -42.917        | -53.8637       | 182.917                 | 133.8637 |
|                        |                         | $\Sigma L = 0$ | $\Sigma D = 0$ | 140 (OK)                | 80 (OK)  |

Using transit rule,

$$\Delta L = e_L \times \frac{L_i}{\Sigma L_i}$$

$$= \cancel{-0.0634}$$

$$= -0.051 \times \frac{L_i}{301.287}$$

$$\Delta D = e_D \times \frac{D_i}{\Sigma D_i}$$

$$= 0.009 \times \frac{D_i}{283.715}$$

Group no: D

Department of Civil Engineering  
BCE Survey Camp - 2080 (2078 Batch)

Observer : Kumar Bhandari  
Recorder : Laxmi Bhatta  
Instrument : Total Station

Detailing by Total Station (Angle and Distance System)

Date : 2080-12-16

|                   |   |   |   |
|-------------------|---|---|---|
| Coordinate:       | N | E | Z |
| Backsight:        |   |   |   |
| Occupied Station: |   |   |   |

Weather:  
Temperature:

| Ins. St. & Ht. of ins. | Sighted to      | Signal height (m) | HCR Observation |    |    | Horizontal Distance (m) | ± | Vertical Distance (m) | Remarks         |
|------------------------|-----------------|-------------------|-----------------|----|----|-------------------------|---|-----------------------|-----------------|
|                        |                 |                   | d               | m  | s  |                         |   |                       |                 |
| D <sub>1</sub>         | D <sub>6</sub>  |                   | 0               | 0  | 0  | 68.830                  | + | 0.086                 |                 |
|                        | FP <sub>2</sub> |                   | 92              | 49 | 46 | 8.935                   | + | 0.376                 | Electric Pole   |
|                        | FP <sub>1</sub> |                   | 100             | 51 | 45 | 10.386                  | + | 0.307                 | " " "           |
|                        | BL <sub>1</sub> |                   | 124             | 40 | 20 | 12.456                  | + | 0.070                 | Boundary line   |
|                        | G <sub>2</sub>  |                   | 73              | 03 | 26 | 24.075                  | + | 0.304                 | Gate            |
|                        | G <sub>1</sub>  |                   | 65              | 34 | 44 | 29.144                  | + | 0.284                 | Gate            |
|                        | b <sub>3</sub>  |                   | 98              | 10 | 53 | 18.884                  | + | 0.515                 | Building corner |
|                        | b <sub>2</sub>  |                   | 106             | 27 | 29 | 27.475                  | + | 0.338                 | " "             |
|                        | R <sub>2</sub>  |                   | 43              | 15 | 36 | 29.055                  | + | 0.388                 | Road edge       |
|                        | R <sub>1</sub>  |                   | 58              | 03 | 42 | 30.757                  | + | 0.342                 | " "             |
|                        | R <sub>3</sub>  |                   | 78              | 02 | 49 | 12.186                  | + | 0.367                 | " "             |
|                        | R <sub>4</sub>  |                   | 52              | 06 | 14 | 9.399                   | + | 0.352                 | " "             |
|                        | BB <sub>1</sub> |                   | 34              | 39 | 21 | 44.500                  | + | 0.064                 | Building        |
|                        | BB <sub>3</sub> |                   | 30              | 09 | 10 | 44.512                  | + | 0.083                 | " "             |
|                        | P <sub>3</sub>  |                   | 50              | 39 | 12 | 29.818                  | + | 0.408                 | Parking         |
|                        | P <sub>1</sub>  |                   | 58              | 41 | 38 | 31.028                  | + | 0.346                 | Parking         |
|                        | B <sub>3</sub>  |                   | 63              | 59 | 27 | 47.461                  | + | 0.519                 | Building        |
|                        | B <sub>2</sub>  |                   | 68              | 11 | 45 | 49.193                  | + | 1.013                 | " "             |
|                        | B <sub>5</sub>  |                   | 56              | 08 | 56 | 49.232                  | + | 0.992                 | " "             |
| D <sub>2</sub>         | D <sub>1</sub>  |                   | 0               | 0  | 0  | 81.395                  | + | 0.111                 |                 |
|                        | b <sub>1</sub>  |                   | 21              | 29 | 13 | 66.674                  | + | 0.341                 | Building        |
|                        | B <sub>1</sub>  |                   | 30              | 04 | 49 | 74.796                  | + | 0.0382                | " "             |
|                        | B <sub>6</sub>  |                   | 35              | 24 | 32 | 69.165                  | + | 0.0357                | " "             |
|                        | B <sub>7</sub>  |                   | 40              | 41 | 55 | 65.165                  | + | 1.995                 | " "             |
|                        | B <sub>8</sub>  |                   | 42              | 10 | 23 | 50.766                  | + | 1.995                 | " "             |
|                        | B <sub>9</sub>  |                   | 39              | 54 | 38 | 58.605                  | + | 1.947                 | " "             |
|                        | B <sub>10</sub> |                   | 38              | 24 | 02 | 64.404                  | + | 1.916                 | " "             |
|                        | B <sub>11</sub> |                   | 56              | 41 | 36 | 49.101                  | + | 0.945                 | " "             |
|                        | B <sub>12</sub> |                   | 64              | 53 | 55 | 48.108                  | + | 0.911                 | " "             |
|                        | B <sub>13</sub> |                   | 67              | 26 | 17 | 51.987                  | + | 0.779                 | " "             |
|                        | B <sub>14</sub> |                   | 78              | 46 | 49 | 27.833                  | + | 0.032                 | Boundary        |

Signature of Teacher at site.....

Page No: 1

Group no: D

**Department of Civil Engineering**  
**BCE Survey Camp - 2080 (2078 Batch)**

Observer : Kumar Detalling by Total Station (Angle and Distance System) Date : 2080-12-16

Recorder : Laxmi Bhatta

Instrument : T.S.

|                   |   |   |   |
|-------------------|---|---|---|
| Coordinate:       | N | E | Z |
| Backsight:        |   |   |   |
| Occupied Station: |   |   |   |

Weather:

Temperature:

| Ins. St. & Ht. of ins. | Sighted to   | Signal height (m) | HCR Observation |    |    | Horizontal Distance (m) | ± | Vertical Distance (m) | Remarks           |
|------------------------|--------------|-------------------|-----------------|----|----|-------------------------|---|-----------------------|-------------------|
|                        |              |                   | d               | m  | s  |                         |   |                       |                   |
| D3                     | D2           |                   | 0               | 0  | 0  | 71.588                  | + | 0.626                 |                   |
|                        | WB1          |                   | 172             | 07 | 28 | 40.389                  | + | 0.043                 | Workshop Building |
|                        | WB2          |                   | 180             | 25 | 17 | 42.204                  | - | -0.177                | " "               |
|                        | WB4          |                   | 169             | 14 | 22 | 52.112                  | - | -0.132                | " "               |
| D4                     | D3           |                   | 0               | 0  | 0  | 74.448                  | + | 0.276                 |                   |
|                        | B12          |                   | 28              | 14 | 03 | 24.469                  | + | 0.399                 | Boundary Building |
|                        | B15          |                   | 33              | 30 | 33 | 44.572                  | + | 1.359                 | " "               |
|                        | B16          |                   | 37              | 07 | 52 | 41.155                  | + | 1.383                 | " "               |
|                        | B17          |                   | 49              | 43 | 41 | 44.119                  | + | 1.462                 | " "               |
|                        | B18          |                   | 53              | 27 | 34 | 40.739                  | + | 1.402                 | " "               |
|                        | B19          |                   | 62              | 22 | 17 | 38.923                  | + | 1.434                 | " "               |
|                        | B20          |                   | 72              | 43 | 41 | 46.600                  | + | 1.442                 | " "               |
|                        | B21          |                   | 76              | 26 | 14 | 59.307                  | + | 1.624                 | " "               |
|                        | B22          |                   | 80              | 49 | 09 | 66.359                  | + | 1.636                 | " "               |
| D5                     | D4           |                   | 0               | 0  | 0  | 66.299                  | - | -0.066                |                   |
|                        | 1            |                   | 221             | 57 | 27 | 10.685                  | + | 0.130                 | Building          |
|                        | 2            |                   | 172             | 34 | 25 | 3.668                   | + | 0.320                 | " "               |
|                        | 3            |                   | 153             | 50 | 35 | 20.225                  | + | 0.562                 | " "               |
|                        | R5           |                   | 130             | 09 | 55 | 25.482                  | + | 0.479                 | Road              |
|                        | R6           |                   | 130             | 18 | 57 | 30.078                  | + | 0.481                 | " "               |
| D6                     | D5           |                   | 0               | 0  | 0  | 83.762                  | - | -0.417                |                   |
|                        | CSIT-1       |                   | 339             | 15 | 50 | 48.256                  | + | 0.437                 | CSIT-1 Building   |
|                        | CSIT-2       |                   | 332             | 17 | 34 | 44.604                  | + | 0.571                 | CSIT-2            |
|                        | CSIT-3       |                   | 326             | 59 | 36 | 53.410                  | + | 1.041                 | CSIT-3            |
|                        | CSIT-4       |                   | 311             | 37 | 10 | 39.205                  | + | 1.011                 | CSIT-4            |
|                        | CSIT-5       |                   | 301             | 45 | 06 | 38.574                  | + | 0.509                 | CSIT-5            |
|                        | CSIT-6       |                   | 301             | 39 | 03 | 54.952                  | + | 0.713                 | CSIT-6            |
|                        | Science Bldg |                   | 292             | 03 | 16 | 25.652                  | + | 0.124                 | Science Bldg      |

Signature of Teacher at site.....

Page No: 2

**Group no:** D

**Department of Civil Engineering**  
**BCE Survey Camp - 2080 (2078 Batch)**

Observer : Kumar Detailing by Total Station (Angle and Distance System) Date : 2020-12-16<sup>th</sup>

Recorder : Laxmi Bhatto

|             |   |   |   |
|-------------|---|---|---|
| Coordinate: | N | E | Z |
|-------------|---|---|---|

**Weather:**

Instrument : Total Station

|                   |
|-------------------|
| Backsight:        |
| Occupied Station: |

**N**

**E**

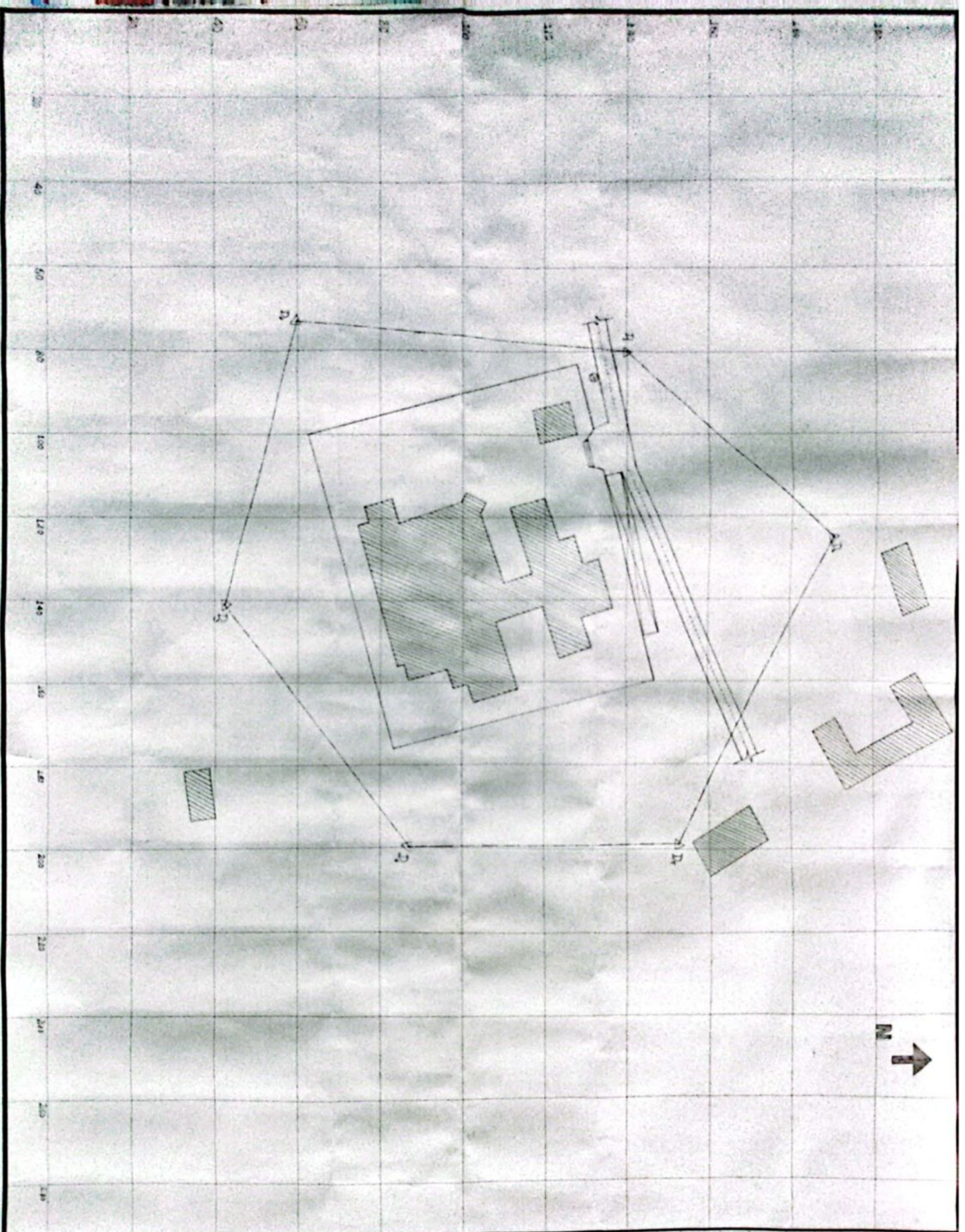
2

**Temperature:**

[illegible]

Signature of Teacher at site.....

Page No: 3



INDEX

| SYMBOL | NAME               |
|--------|--------------------|
|        | GATE               |
|        | ELECTRICAL POLE    |
|        | ROAD               |
|        | PARKING            |
|        | BUILDING           |
|        | GROUND POINT       |
|        | INSTRUMENT STATION |
|        | BOUNDARY LINE      |
|        | TWICE LINE         |

SCALE 1:400

*Handwritten signature*

DRAWN BY: [illegible]  
 CHECKED BY: [illegible]  
 DATE: [illegible]  
 PROJECT: [illegible]

## PRACTICAL-2

TITLE = TACHEOMETRY

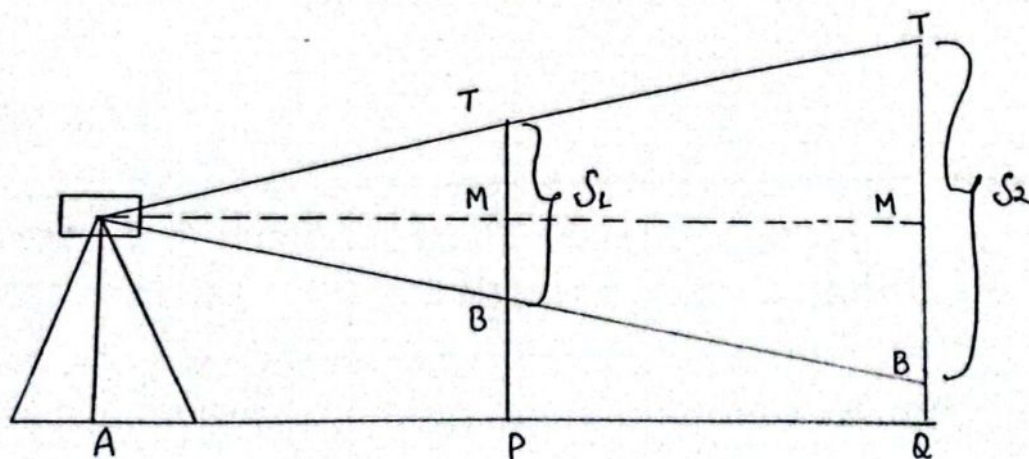
OBJECTIVE :- TO DETERMINE THE VALUE OF STADIA INTERVAL FACTOR ( $K$ ) AND STADIA CONSTANT ( $C$ ) OF THE INSTRUMENT.

INSTRUMENT REQUIRED :-

- ① Theodolite
- ② Pegs
- ③ Tape
- ④ staff
- ⑤ Hammer
- ⑥ Tripod.

THEORY :-

A branch of angular survey in which both, horizontal and vertical positions of points are determined from the horizontal instrumental observation is called tacheometry. The instrument employed for measuring angular readings in tacheometric survey is termed as tachometer. The accuracy obtained in tacheometry is 1:1000 for favourable condition.



## PROCEDURE:-

1. Fairly levelled ground was selected.
2. Theodolite was setup at A and at known distance, Point Q was selected.
3. Another point P was selected at approximately bet<sup>n</sup> A & B
4. Then the staff was holded at station P and Q and staff intercept  $s_1$  &  $s_2$  were taken respectively.
5. All the process were done by making telescope as horizontal so, the horizontal distance formula for horizontal line of sight is,

$$D = KS + C \quad \text{--- (i)}$$

from figure,

$$D_1 = KS_1 + C \quad \text{--- (ii)}$$

$$\text{? } D_2 = KS_2 + C \quad \text{--- (iii)}$$

from (ii) & (iii)

$$K = \frac{D_2 - D_1}{S_2 - S_1} \quad \text{--- (iv)}$$

? from eqn (ii)

$$C = \frac{D_1 S_2 - D_2 S_1}{S_2 - S_1} \quad \text{--- (v)}$$

## OBSERVATION:-

| Instrument station | Sighted to | Horizontal distance | Staff reading (m)   |
|--------------------|------------|---------------------|---------------------|
| A                  | P          | 30m                 | 1.445, 1.295, 1.145 |
|                    | Q          | 50m                 | 1.36, 1.21, 0.86    |

### CALCULATION:-

$$S_1 = 1.445 - 1.145 = 0.3$$

$$S_2 = 1.36 - 0.86 = 0.5$$

$$D_1 = 30\text{m}$$

$$D_2 = 50\text{m}$$

Using eqn (iv)

$$K = \frac{D_2 - D_1}{S_2 - S_1} = \frac{50 - 30}{0.5 - 0.3}$$

$$K = 100$$

Using eqn (v)

$$C = \frac{D_1 S_2 - D_2 S_1}{S_2 - S_1}$$
$$= \frac{30 \times 0.5 - 50 \times 0.3}{0.5 - 0.3}$$

$$C = 0$$

### RESULT:-

The value of  $K$  &  $C$  obtained is 100 & 0 respectively.

## PRACTICAL:-

TITLE :- TRIGONOMETRY LEVELLING

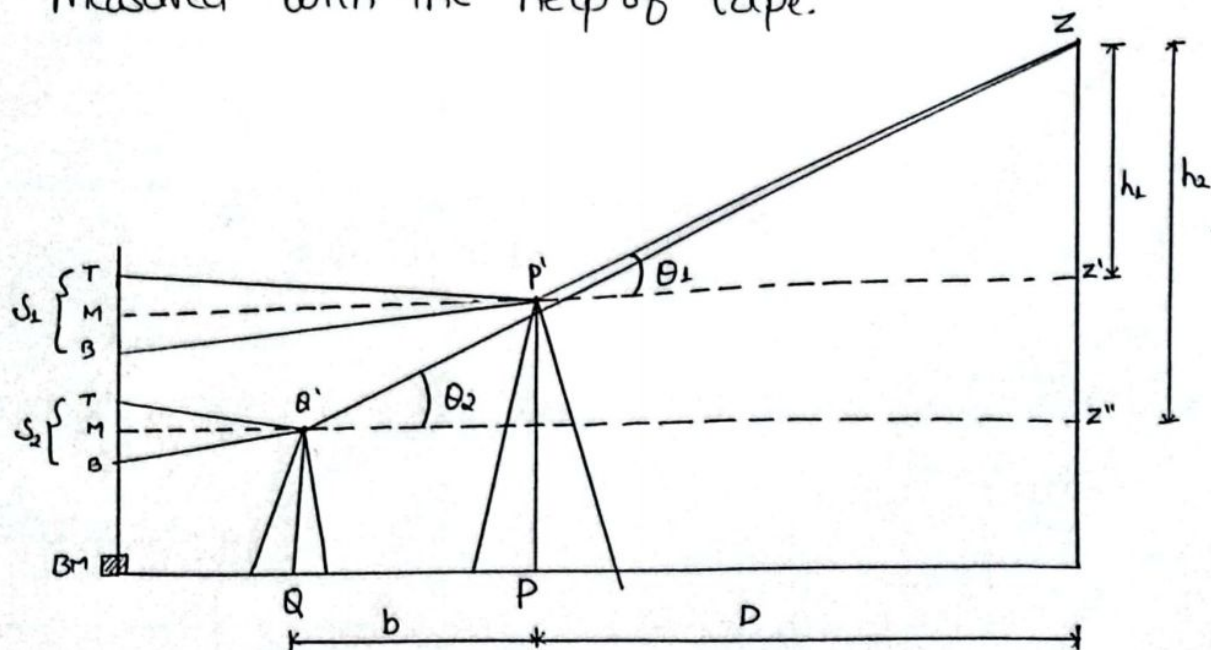
OBJECTIVE:- DETERMINATION OF ELEVATION OF POINT BY TRIGONOMETRIC LEVELLING WHEN INSTRUMENT STATIONS ARE IN SAME VERTICAL PLANE.

INSTRUMENT REQUIRED:-

Theodolite, Tripod, Peg, Hammer, Tape, Staff.

THEORY:-

Trigonometric levelling is the process of determining the difference of elevation of station from observed vertical angles and known distances, which are assumed to be either horizontal or geodetic length of mean sea level. The vertical angles may be measured by means of an accurate theodolite & the horizontal distances are measured with the help of tape.



PROCEDURE:-

- i) Theodolite was setup at P & levelled accurately with respect to altitude bubble.
- ii) With the Vertical Vernier set to zero, & the altitude bubble in the center of the run, reading on the staff was taken which was kept at nearby B.M.

- iii) Then the telescope was directed toward z and bisected accurately. Both the planes were clamped & vertical angle  $\theta_1$  was observed.
- iv) The instrument was shifted to Q & levelled there. And by similar process as above, vertical angle  $\theta_2$  was observed with both face observation.
- v) The distance PQ was measured by tape.

From  $\Delta zp'z'$

$$\tan \theta_1 = \frac{h_1}{D}$$

$$h_1 = D \tan \theta_1 \quad \text{--- (i)}$$

From  $\Delta zQ'z''$

$$\tan \theta_2 = \frac{h_2}{D+b}$$

$$h_2 = (D+b) \tan \theta_2 \quad \text{--- (ii)}$$

Subtracting eqn (ii) from (i)

$$h_2 - h_1 = (D+b) \tan \theta_2 - D \tan \theta_1$$

$$S = D (\tan \theta_2 - \tan \theta_1) + b \tan \theta_2$$

$$D = \frac{S - b \tan \theta_2}{\tan \theta_2 - \tan \theta_1} \quad \text{--- (iii)}$$

From eqn (i)

$$h_1 = \frac{(S - b \tan \theta_2) \cdot \tan \theta_1}{\tan \theta_2 - \tan \theta_1} \quad \text{--- (iv)}$$

OBSERVATION:-

| Instrument Station | Reading on staff kept on BM | Vertical angle |
|--------------------|-----------------------------|----------------|
| P                  | 124.7 m                     | 5° 46' 48"     |
| Q                  | 112.6 m                     | 5° 43' 52"     |

### CALCULATION:-

$$S = 1.247 - 1.126 = 0.121$$

$$\theta_1 = 5^\circ 46' 48''$$

$$\theta_2 = 5^\circ 43' 52''$$

$$b = 4.33\text{m}$$

Now Using eq<sup>n</sup> (iii)

$$D = \frac{S - b \tan \theta_2}{\tan \theta_2 - \tan \theta_1} = \frac{0.121 - 4.33 \times \tan(5^\circ 43' 52'')}{\tan(5^\circ 43' 52'') - \tan(5^\circ 46' 48'')}$$

$$D = 363.79\text{m}$$

Using eq<sup>n</sup> (i)

$$\begin{aligned} h_1 &= D \tan \theta_1 \\ &= 363.79 \times \tan(5^\circ 46' 48'') \\ &= 36.824 \end{aligned}$$

$$\text{let, R.L. of BM} = 200\text{m}$$

$$\begin{aligned} \text{Then, R.L. of Z} &= \text{R.L. of BM} + S + h_1 \\ &= 200 + 0.121 + 1.247 + 36.824 \\ &= 238.071 \end{aligned}$$

### RESULT:-

Hence the R.L. of Z was found to be 238.071m

## PRACTICAL NO:

TITLE:- CURVE SETTING

OBJECT:- TO SET OUT A SIMPLE CIRCULAR CURVE BY RANKINE / ONE THEODOLITE METHOD.

INSTRUMENT REQUIRED:-

Theodolite, tape, pegs, Ranging rods etc.

THEORY:-

The angle between back tangent of Curve joining the point of Commencement Curve and the other end of Curve is called deflection angle. In this method, the Curve is set out using deflection angles and their respective chord length.

The deflection angles by Rankine method are given by

$$\delta_1 = 90 \times \frac{C_1}{\pi R}$$

$$\delta_2 = 90 \times \frac{C_2}{\pi R} = \delta_3 = \delta_4 \dots \dots = \delta_{n-1} = \delta_n$$

And,

$$\Delta_1 = \delta_1$$

$$\Delta_2 = \Delta_1 + \delta_2$$

$$\vdots \quad \vdots \quad \vdots$$

$$\Delta_n = \Delta_{n-1} + \delta_n$$

PROCEDURE:-

- i) Tangent points  $T_1$  &  $T_2$  was located.
- ii) Theodolite was adjusted temporarily on Tangent Point  $T_1$ .
- iii) Using horizontal angle, the angle  $\Delta_1$  was rotated towards the ranging rod which is at a distance equal to the length of first sub chord from  $T_1$ .

iv) Theodolite was then rotated by  $\Delta_2$  about vertical axis toward F which is at a distance equal to the length of normal Chord from D.

v) The process (i), (ii), (iii), (iv) was repeated until the point  $T_2$  is set out are fixed.

### CALCULATION:-

Chainage of IP = 7+486m

$$\Delta = 36^\circ 0' 0''$$

$$R = 100\text{m}$$

Peg Interval = 10m

Now,

$$\begin{aligned}\text{Tangent length (T)} &= R \tan \frac{\Delta}{2} = 100 \times \tan(18^\circ) \\ &= 32.49\text{m}\end{aligned}$$

$$\begin{aligned}\text{Chainage of } T_1 &= 7+486 + 32.49 \\ &= 7+453.51\text{m}\end{aligned}$$

$$\text{Length of Curve} = \frac{\pi R \Delta}{180^\circ} = \frac{\pi \times 100 \times 36}{180^\circ} = 62.83\text{m}$$

$$\begin{aligned}\text{Chainage of Tangency } T_2 &= 7+453.51 + 62.83 \\ &= 7+516.34\text{m}\end{aligned}$$

length of first sub chord ( $C_1$ ) = 6.49m

length of normal chord C = 10m

length of last sub chord  $C_7$  = 6.34m

Tangent angle :-

$$\delta_1 = \frac{90 \times C_1}{\pi R} = \frac{90 \times 6.49}{\pi \times 100} = 1^\circ 51' 33''$$

$$\delta_2 = \delta_3 = \delta_4 = \delta_5 = \delta_6 = \frac{90 \times C}{\pi R} = \frac{90 \times 10}{\pi \times 100} = 2^\circ 51' 53''$$

$$\delta_7 = \frac{90 \times C_7}{\pi R} = \frac{90 \times 6.34}{\pi \times 100} = 1^\circ 48' 59''$$

| Chord No | Chainage | Chord length | Tangential angle | Deflection angle | Setting angle<br>L.C. of the 20" |
|----------|----------|--------------|------------------|------------------|----------------------------------|
| 1        | 7+460    | 6.49         | 1° 51' 33"       | 1° 51' 33"       | 1° 51' 40"                       |
| 2        | 7+470    | 10           | 2° 51' 33"       | 4° 43' 26"       | 4° 43' 20"                       |
| 3        | 7+480    | 10           | 2° 51' 33"       | 7° 35' 20"       | 7° 35' 20"                       |
| 4        | 7+490    | 10           | 2° 51' 33"       | 10° 27' 20"      | 10° 27' 20"                      |
| 5        | 7+500    | 10           | 2° 51' 33"       | 13° 19' 07"      | 13° 19' 00"                      |
| 6        | 7+510    | 10           | 2° 51' 33"       | 16° 11' 00"      | 16° 11' 00"                      |
| 7        | 7+51634  | 6.34         | 1° 48' 59"       | 17° 59' 59"      | 18° 0' 0"                        |

For check  $A_2 = \frac{36^\circ}{2} = 18^\circ$  Checked

## PRACTICAL :-

TITLE:- INDIRECT CONTOURING

OBJECTIVE :- TO PLOT CONTOUR BY SQUARE METHOD.

INSTRUMENT REQUIRED:-

Level, levelling staff, Tape, Ranging rod, Pegs, Hammer.

THEORY:-

INDIRECT METHOD:-

In this method, points are located in the field, generally as corners of well shape geometrical figures such as squares, rectangles etc. Grid/Square method is one of the indirect method of Contouring. In grid method, the area to be surveyed is divided into a grid of squares which may vary from  $5m \times 5m$  to  $25m \times 25m$  depending upon nature of terrain. Contour interval required is scale of map desired. This method is suitable for large scale mapping and at average precision.

PROCEDURE:-

- i) A square of  $25m \times 25m$  was found using tape on the area to be surveyed. On the corner of square, pegs were driven. The square was properly checked using 3-4-5 rule & also by measuring its diagonal.
- ii) The square was further divided into 5 parts both in rows & column using tape. The intersection point of row and column were marked by driving peg in it.
- iii) Hence, each square formed inside the area of  $25 \times 25m$  was  $5m \times 5m$ .

4. In this way, the area for Contouring was marked.
5. Then instrument was set on point 'p' (say) from where all the points on square were clearly visible, i.e. instrument st<sup>n</sup> was so selected such that staff reading could be taken easily.
6. First the staff reading was taken on assumed BM. of R.L. 200m. Hence, height of instrument was found by adding R.L. of BM & its staff reading (B.S) . i.e.

$$H.I = R.L \text{ of BM} + BS = 200 + B.S.$$

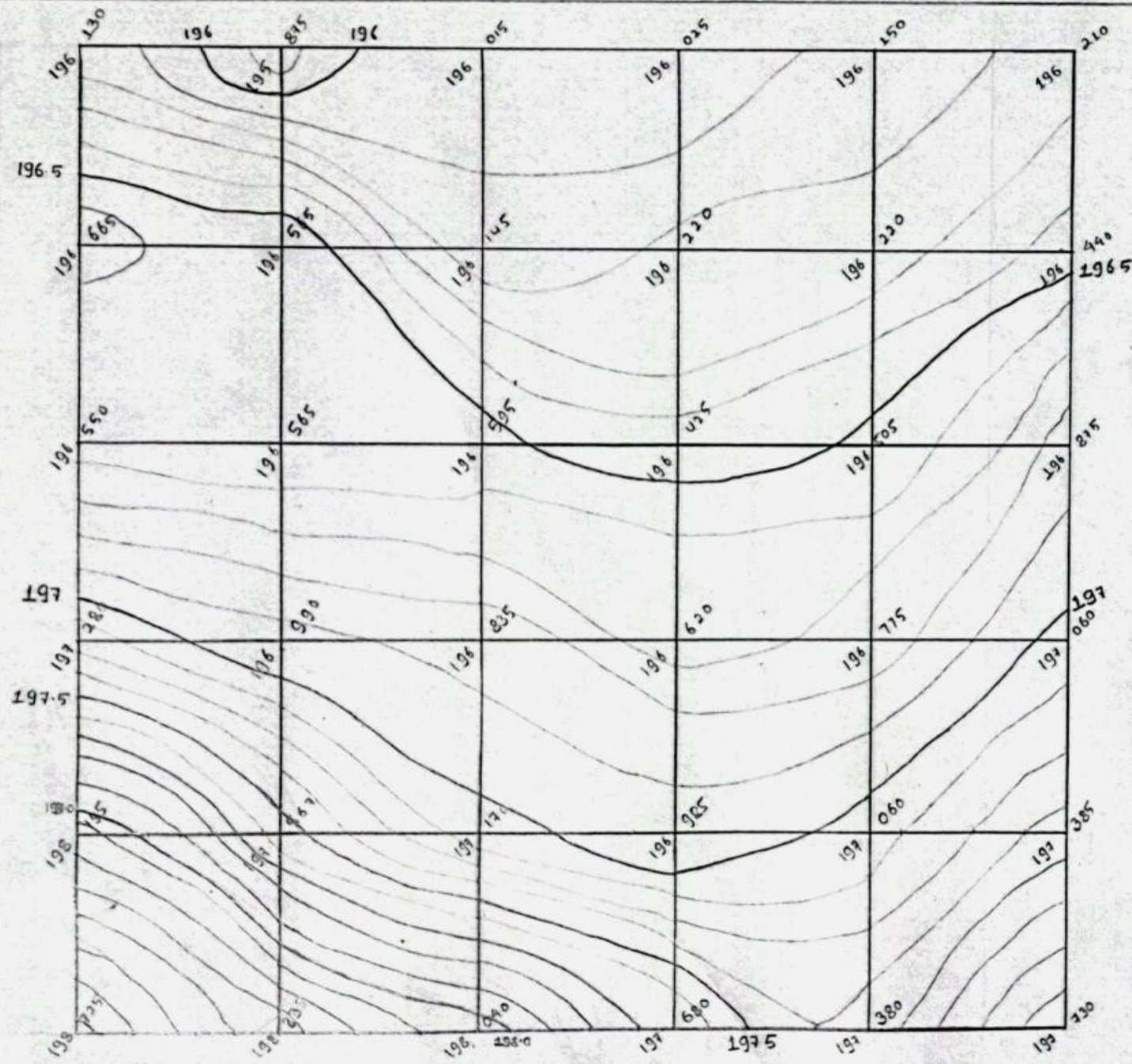
7. Then staff reading on each point on square were taken. Thus, R.L. of point on square was calculated by subtracting staff reading of that point from H.I.

$$\text{i.e. R.L. of point} = H.I - \text{Staff reading of that point.}$$

8. Contours were ~~taken~~ then drawn on paper with Contour interval of 0.1m.

| S.N. | B.S.  | I.S.  | F.S.  | Rise / Fall | R.L. (m) | Remarks               |
|------|-------|-------|-------|-------------|----------|-----------------------|
| 1    | 0.84  |       |       |             | 200      | TBM                   |
| 2    | 0.495 |       | 2.780 | -1.940      | 198.06   | C.P-1                 |
| 3    |       | 2.345 |       | -1.850      | 196.210  | A <sub>6</sub>        |
| 4    |       | 2.425 |       | -1.930      | 196.130  | A <sub>1</sub>        |
| 5    |       | 0.825 |       | -0.330      | 197.730  | F <sub>6</sub>        |
| 6    |       | 2.115 |       | -1.620      | 196.440  | B <sub>6</sub>        |
| 7    |       | 1.680 |       | -1.185      | 196.875  | C <sub>6</sub>        |
| 8    |       | 1.495 |       | -1.000      | 197.060  | D <sub>6</sub>        |
| 9    |       | 1.170 |       | -0.675      | 197.385  | E <sub>6</sub>        |
| 10   |       | 2.405 |       | -1.910      | 196.150  | A <sub>5</sub>        |
| 11   |       | 2.530 |       | -2.035      | 196.025  | A <sub>4</sub>        |
| 12   |       | 2.540 |       | -2.045      | 196.015  | A <sub>3</sub>        |
| 13   |       | 2.680 |       | -2.185      | 195.875  | A <sub>2</sub>        |
| 14   |       | 1.175 |       | -0.680      | 197.380  | F <sub>5</sub>        |
| 15   |       | 0.875 |       | -0.380      | 197.680  | F <sub>4</sub>        |
| 16   |       | 0.515 |       | -0.020      | 198.040  | F <sub>3</sub>        |
| 17   |       | 0.220 |       | 0.275       | 198.335  | F <sub>2</sub>        |
| 18   |       | 1.495 |       | -1.000      | 197.060  | E <sub>5</sub>        |
| 19   |       | 1.78  |       | -1.285      | 196.775  | D <sub>5</sub>        |
| 20   |       | 2.050 |       | -1.555      | 196.505  | C <sub>5</sub>        |
| 21   | 1.520 |       | 2.345 | -1.850      | 196.210  | A <sub>1</sub> /C.P-2 |
| 22   | '     | 1.510 |       | 0.010       | 196.220  | B <sub>5</sub>        |
| 23   |       | 1.510 |       | 0.010       | 196.220  | B <sub>4</sub>        |
| 24   |       | 1.305 |       | 0.215       | 196.425  | C <sub>4</sub>        |
| 25   |       | 1.110 |       | 0.410       | 196.620  | D <sub>4</sub>        |
| 26   |       | 0.745 |       | 0.775       | 196.985  | E <sub>4</sub>        |
| 27   |       | 1.585 |       | -0.065      | 196.145  | B <sub>3</sub>        |
| 28   |       | 1.135 |       | 0.385       | 196.595  | C <sub>3</sub>        |
| 29   |       | 0.895 |       | 0.625       | 196.835  | D <sub>3</sub>        |
| 30   |       | 0.740 |       | 0.780       | 196.990  | D <sub>2</sub>        |
| 31   |       | 1.165 |       | 0.355       | 196.565  | C <sub>2</sub>        |
| 32   |       | 2.165 |       | -0.645      | 196.565  | B <sub>2</sub>        |
| 33   |       | 2.065 |       | -0.545      | 196.665  | B <sub>1</sub>        |
| 34   |       | 1.18  |       | 0.340       | 196.550  | C <sub>1</sub>        |
| 35   | 2.045 |       | 0.45  | 1.070       | 197.280  | D <sub>1</sub> /C.P-3 |





SCALE = 1:100

FIG :- CONTOUR MAP  
BY GRID METHOD

## PRACTICAL -

TITLE :- INDIRECT METHOD OF CONTOURING [RADIAL]

OBJECT :- TO PREPARE A CONTOUR MAP OF THE GIVEN AREA BY RADIAL LINE METHOD.

INSTRUMENT REQUIRED :-

Theodolite, prism, ranging rod, pegs, Tripod etc.

THEORY :-

In this method, the area to be surveyed is divided into a number of equal angular intervals about a common point. The points which lie on each line are at a distance depending upon the slope of the ground. Points are taken at smaller distance at steep ground & at greater distance on mild slope. It can be achieved by using either theodolite or total station. Total station with prism gives horizontal & vertical distances directly whereas theodolite is used with levelling staff & by using tachemetry, horizontal distance (D) & vertical distance (V) are found out as,

$$D = KS (\cos^2 \theta + C \cos \theta)$$

$$V = \frac{1}{2} KS \sin 2\theta + C \sin \theta$$

This method is cheaper and less tedious as compared to direct method but also less accurate. Thus it can be used when time & budget is limited.

The contour map is then prepared by interpolating the points of equal R.L. The R.L. of any point is given

$$R.L. = R.L. \text{ of BM} + H.I. \pm V$$

This method is also called spot method of contouring.

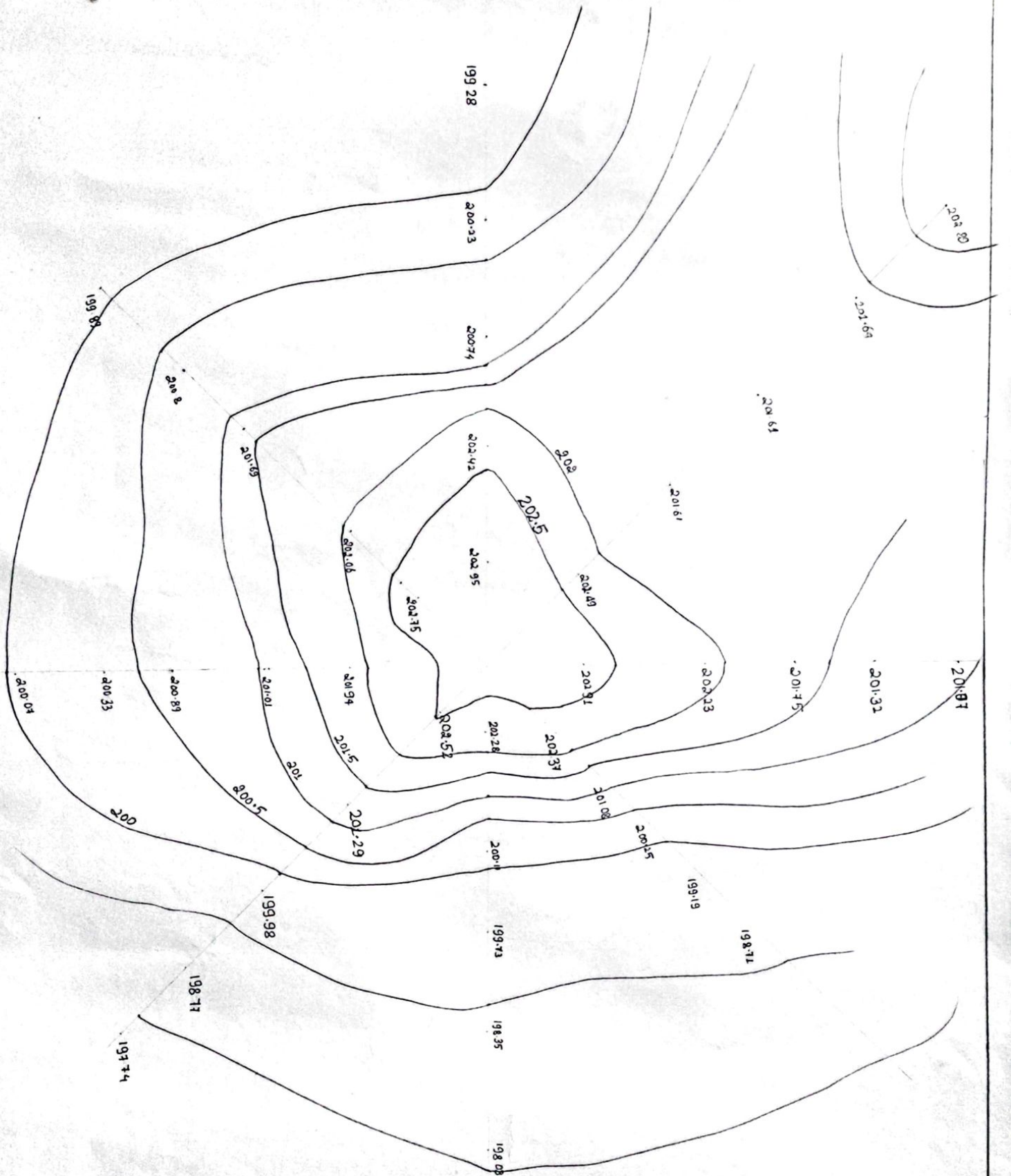
## PROCEDURE:-

- i) A B.M. was selected & the instrument was adjusted on it. The H.I. was measured.
- ii) An initial line was taken & pegs were driven along that line at an interval of 5-10m while keeping the horizontal angle at  $0^{\circ}0'0''$ .
- iii) The same procedure was repeated after rotating the instrument along vertical axis by  $45^{\circ}0'0''$ .
- iv) Horizontal & vertical distances were measured by using prism at each point on a single radial line.
- v) At angular interval by  $45^{\circ}0'0''$  the step ii & step iv were repeated until full turn ( $360^{\circ}$ ) was completed.

| STN | SIGHTED TO | HCR                  | VERTICAL ANGLE         | STAFF READING            |       |       | STAFF INTERCEPT           | HORIZONTAL DISTANCE      | VERTICAL DISTANCE         | R.L.                        |
|-----|------------|----------------------|------------------------|--------------------------|-------|-------|---------------------------|--------------------------|---------------------------|-----------------------------|
|     |            |                      |                        | T                        | M     | B     |                           |                          |                           |                             |
| A   | BM         | <del>85° 0' 0"</del> | <del>92° 38' 40"</del> | <del>1.954</del><br>1.58 | 1.58  | 1.215 | <del>0.739</del><br>0.055 | <del>86.73</del><br>5.44 | <del>-3.43</del><br>-0.52 | <del>203.43</del><br>202.91 |
|     | 1          | 0° 0' 0"             | 95° 32' 19"            | 1.610                    | 1.585 | 1.555 | 0.055                     | 5.44                     | -0.52                     | 202.91                      |
|     | 2          | 0° 0' 0"             | 95° 32' 19"            | 2.030                    | 1.965 | 1.905 | 0.125                     | 12.38                    | -1.20                     | 202.23                      |
|     | 3          | 0° 0' 0"             | 95° 32' 19"            | 1.530                    | 1.445 | 1.355 | 0.175                     | 17.33                    | -1.68                     | 201.75                      |
|     | 4          | 0° 0' 0"             | 95° 32' 19"            | 0.860                    | 0.750 | 0.640 | 0.220                     | 21.79                    | -2.11                     | 201.32                      |
|     | 5          | 0° 0' 0"             | 95° 7' 59"             | 0.900                    | 0.770 | 0.635 | 0.265                     | 26.28                    | -2.36                     | 201.07                      |
|     | 1          | <del>45° 0' 0"</del> | 104° 4' 0"             | 1.175                    | 1.150 | 1.130 | 0.045                     | 4.23                     | -1.06                     | 202.37                      |
|     | 2          | 45° 0' 0"            | 104° 3' 57"            | 1.200                    | 1.150 | 1.100 | 0.100                     | 9.40                     | -2.35                     | 201.08                      |
|     | 3          | 45° 0' 0"            | 104° 3' 51"            | 0.765                    | 0.700 | 0.630 | <del>0.630</del><br>0.180 | 12.70                    | -3.18                     | <del>200.95</del><br>199.19 |
|     | 4          | 45° 0' 0"            | 104° 3' 51"            | 0.395                    | 0.305 | 0.215 | 0.180                     | 16.93                    | -4.24                     | 199.19                      |
|     | 5          | 45° 0' 0"            | 102° 43' 87"           | 1.025                    | 0.915 | 0.805 | 0.220                     | 20.93                    | -4.72                     | 198.71                      |
|     | 1          | 90° 0' 0"            | 107° 39' 39"           | 1.230                    | 1.210 | 1.190 | 0.040                     | 3.63                     | -1.15                     | 202.28                      |
|     | 2          | 90° 0' 0"            | 107° 39' 53"           | 0.625                    | 0.570 | 0.510 | 0.115                     | 10.44                    | -3.32                     | 200.11                      |
|     | 3          | 90° 0' 0"            | 103° 46' 42"           | 1.160                    | 1.080 | 1.000 | 0.160                     | 15.09                    | -3.70                     | 199.73                      |
|     | 4          | 90° 0' 0"            | 103° 46' 42"           | 0.685                    | 0.570 | 0.465 | 0.220                     | 20.75                    | -5.08                     | 198.35                      |
|     | 5          | 90° 0' 0"            | 101° 8' 17"            | 1.150                    | 1.005 | 0.865 | 0.285                     | 27.43                    | -5.40                     | 198.03                      |
|     | 1          | 135° 0' 0"           | 103° 35' 21"           | 1.320                    | 1.285 | 1.280 | 0.040                     | 3.77                     | -0.91                     | 202.52                      |
|     | 2          | 135° 0' 0"           | 100° 57' 05"           | 1.975                    | 1.920 | 1.860 | 0.115                     | 11.08                    | -2.41                     | 201.29                      |
|     | 3          | 135° 0' 0"           | 100° 57' 54"           | 1.805                    | 1.710 | 1.620 | 0.185                     | 17.83                    | -3.45                     | 199.98                      |
|     | 4          | 135° 0' 0"           | 100° 57' 54"           | 1.535                    | 1.410 | 1.285 | 0.250                     | 24.09                    | -4.66                     | 198.77                      |
|     | 5          | 135° 0' 0"           | 100° 57' 58"           | 1.100                    | 0.945 | 0.795 | 0.305                     | 29.39                    | -5.69                     | 197.74                      |

(200m)

| STN | SIGHTED TO | HCR        | VERTICAL ANGLE | STAFF READING |       |       | STAFF INTERCEPT | HORIZONTAL DISTANCE | VERTICAL DISTANCE | R.L.   |
|-----|------------|------------|----------------|---------------|-------|-------|-----------------|---------------------|-------------------|--------|
|     |            |            |                | T             | M     | B     |                 |                     |                   |        |
|     | 1          | 180° 0' 0" | 100° 57' 24"   | 0.585         | 0.545 | 0.505 | 0.080           | 7.71                | -1.49             | 201.94 |
|     | 2          | 180° 0' 0" | 100° 57' 27"   | 0.425         | 0.365 | 0.295 | 0.130           | 12.53               | -2.42             | 201.01 |
|     | 3          | 180° 0' 0" | 98° 12' 42"    | 1.065         | 0.975 | 0.885 | 0.180           | 17.63               | -2.54             | 200.89 |
|     | 4          | 180° 0' 0" | 98° 12' 38"    | 1.020         | 0.910 | 0.800 | 0.220           | 21.55               | -3.10             | 200.33 |
|     | 5          | 180° 0' 0" | 97° 13' 37"    | 1.400         | 1.260 | 1.130 | 0.270           | 26.57               | -3.36             | 200.07 |
|     | 1          | 225° 0' 0" | 97° 13' 26"    | 1.180         | 1.155 | 1.125 | 0.055           | 5.41                | -0.68             | 202.75 |
|     | 2          | 225° 0' 0" | 97° 13' 37"    | 1.985         | 1.900 | 1.875 | 0.110           | 10.82               | -1.37             | 202.06 |
|     | 1          | 270° 0' 0" | 94° 40' 42"    | 1.595         | 1.565 | 1.535 | 0.060           | 5.96                | -0.48             | 202.95 |
|     | 2          | 270° 0' 0" | 94° 40' 42"    | 1.900         | 1.835 | 1.775 | 0.125           | 12.41               | -1.01             | 202.42 |
|     | 3          | 270° 0' 0" | 98° 14' 59"    | 1.075         | 0.980 | 0.885 | 0.190           | 18.60               | -2.69             | 200.74 |
|     | 4          | 270° 0' 0" | 97° 17' 08"    | 1.365         | 1.240 | 1.110 | 0.255           | 25.08               | -3.20             | 200.23 |
|     | 5          | 270° 0' 0" | 97° 17' 21"    | 1.020         | 0.855 | 0.690 | 0.330           | 32.46               | -4.15             | 199.28 |
|     | 1          | 315° 0' 0" | 97° 17' 40"    | 1.385         | 1.345 | 1.310 | 0.075           | 7.37                | -0.94             | 202.49 |
|     | 2          | 315° 0' 0" | 97° 17' 41"    | 1.015         | 0.940 | 0.870 | 0.145           | 14.26               | -1.82             | 201.61 |
|     | 3          | 315° 0' 0" | 94° 53' 42"    | 1.185         | 1.020 | 0.910 | 0.215           | 21.34               | -1.82             | 201.61 |
|     | 4          | 315° 0' 0" | 93° 33' 42"    | 0.740         | 0.600 | 0.450 | 0.290           | 28.88               | -1.79             | 201.64 |
|     | 5          | 315° 0' 0" | 90° 56' 37"    | 1.085         | 0.905 | 0.725 | 0.360           | 35.99               | -0.59             | 202.84 |
|     |            |            |                |               |       |       |                 |                     |                   |        |
|     |            |            |                |               |       |       |                 |                     |                   |        |
|     |            |            |                |               |       |       |                 |                     |                   |        |
|     |            |            |                |               |       |       |                 |                     |                   |        |



SCALE = 1:200